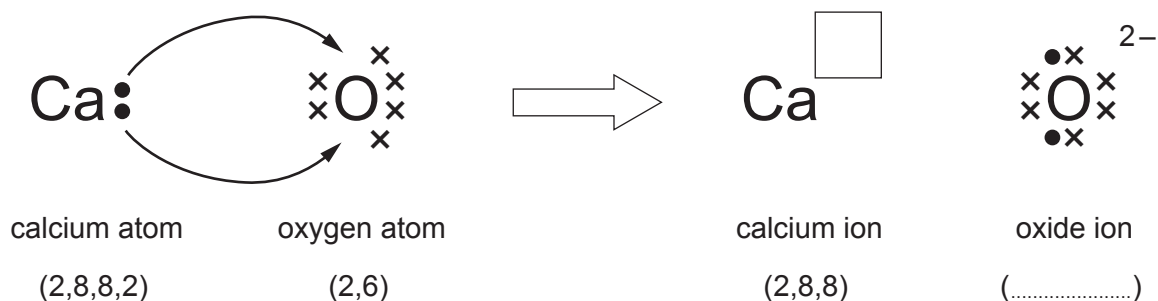


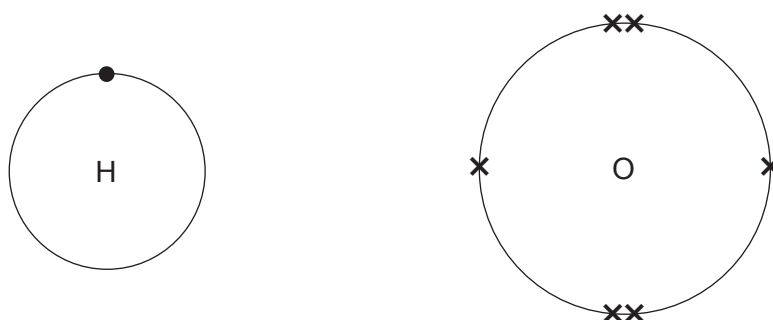
6. (a) The diagram shows the electronic changes that occur when calcium reacts with oxygen to form calcium oxide. The  $\bullet$  and  $\times$  symbols represent outer shell electrons.



- (i) Complete the right-hand side of the diagram by putting the **charge** of the calcium ion in the box and the **electronic structure** of the oxide ion in the brackets. [2]
- (ii) Complete and balance the symbol equation for the reaction between calcium and oxygen. [2]



- (b) The diagrams below show the outer shell electrons in an atom of hydrogen and an atom of oxygen.



Complete the following diagram to show the outer shell electrons in a molecule of water. [2]

